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[Aravindan Natarajan](#) * and [Aravind Kumar Arunagiri](#)

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Article

A Physics-Informed Clustering and Class-Balanced Machine Learning Framework for Industrial Pump Health Monitoring

Aravindan Natarajan * and Aravind Kumar Arunagiri

Independent Researcher, Bangalore, 560109 Karnataka, India

* Correspondence: anatarajank@gmail.com

Abstract

This research presents a multi-stage methodology for analyzing and predicting industrial pump fleet health using sensor data. Following exploratory data analysis, a K-means++ clustering algorithm segmented the data into distinct operational states. Statistical validation confirmed significant variance in variable means across clusters. A physics-informed labeling strategy subsequently mapped these clusters to four domain-specific health states: *Normal*, *Healthy*, *Impending Failure*, and *End of Life*. The resulting dataset was utilized to evaluate multiple classification architectures. The Light Gradient Boosting Machine (LightGBM) framework emerged as the optimal model. Optimized via Bayesian hyperparameter tuning, this architecture successfully countered severe class prevalence asymmetry, yielding high sensitivity for the critical *Impending Failure* minority state label while effectively resolving overlapping physical transition boundaries. The framework achieved robust predictive performance across training and unseen validation datasets. To ensure transparency and foster operator trust, Local Interpretable Model-agnostic Explanations (LIME) were integrated to provide instance-level feature contributions, delineating the specific sensor thresholds triggering health-state predictions. Ultimately, this work demonstrates the efficacy of integrating unsupervised and supervised learning frameworks with domain expertise to deliver robust, explainable predictive maintenance solutions.

Keywords: physics-informed machine learning; predictive maintenance; industrial pumps; sensor data analytics; class imbalance learning; Explainable AI (XAI)

1. Introduction

The reliability of large industrial pumps is a critical factor for maintaining operational continuity across key sectors like manufacturing, oil and gas, water treatment, and power generation [1,2]. Traditional maintenance paradigms, which are either reactive or based on fixed intervals, frequently incur unscheduled downtime, significant operational expenditure, and suboptimal asset performance [1–3]. The emergence of advanced sensor technology and data analytics provides a compelling alternative. By leveraging continuous sensor data to monitor pump health, it is now possible to implement predictive maintenance strategies [4]. This proactive approach allows for the early detection of potential failures, enabling maintenance to be scheduled precisely when needed, which in turn optimizes operational workflows and extends the service life of critical equipment [5].

The field of predictive maintenance for industrial pumps has undergone a significant transformation, driven by advancements in sensor technology, the Industrial Internet of Things (IIoT), and sophisticated machine learning (ML) and deep learning (DL) algorithms [4,5]. This shift has facilitated a move away from traditional reactive and time-based maintenance strategies toward proactive, data-driven approaches that anticipate equipment failures, optimize maintenance schedules, and improve overall operational efficiency and safety [6–8].

Current research on pump health monitoring primarily leverages real-time data from various sensors, including those for vibration, temperature, pressure, flow rate, and motor current signature analysis (MCSA) [8]. This sensor data serves as the foundation for developing intelligent systems capable of early anomaly detection and accurate fault prediction.

A wide array of machine learning techniques is employed in this domain. Supervised learning algorithms such as Support Vector Machines (SVMs), Decision Trees, Random Forests, and XGBoost remain popular for fault classification and diagnosis due to their robust performance and interpretability on structured sensor data [9,10]. These models are commonly trained to classify distinct operational states or specific fault conditions, which allows operators to identify deviations from normal behavior [11–18].

Recent research has shown a growing interest in deep learning approaches, particularly for processing large volumes of time-series sensor data and automatically extracting complex, non-linear features. Convolutional Neural Networks (CNNs) are frequently used to process time-frequency representations of signals, such as spectrograms of vibration data, for fault diagnosis [11]. Recurrent Neural Networks (RNNs), including Long Short-Term Memory (LSTM) networks, are gaining traction for their ability to model temporal dependencies in sensor data, which is crucial for predicting Remaining Useful Life (RUL) and identifying evolving degradation patterns [12]. Unsupervised methods like autoencoders are also used for anomaly detection by learning normal operating conditions and flagging significant deviations without the need for labeled fault data [19].

Furthermore, the integration of data-driven models with physics-based models or domain expertise is an emerging trend [8,9]. These hybrid approaches combine the strengths of both methodologies, offering more robust predictions, especially when data is scarce or when a deeper understanding of underlying degradation mechanisms is required. The use of digital twin technology i.e., creating virtual replicas of physical assets, is also being explored to simulate pump behavior and degradation, enabling more precise predictive maintenance strategies [10].

While significant progress has been made in predictive maintenance for industrial pumps, several key challenges remain [1–3]. These include ensuring the quality and consistency of sensor data, the scarcity of labeled fault data required for supervised learning, and the need for greater interpretability of complex models [5,6]. The lack of labeled data is particularly challenging, as pumps can operate for extended periods without failure, and the data itself is often high-volume and variable in quality [2,11]. To bridge these gaps and address the hurdles of real-world deployment, this research proposes a novel, multi-stage framework. The study is conducted on a synthetic dataset of 20,000 data points from five different pumps. The methodology begins with an exploratory data analysis, followed by the application of clustering techniques to identify and group distinct operational states. These states are then labeled using a physics-informed approach, and their validity is confirmed through statistical analysis. The final stage involves training and evaluating various classification models to predict these states and assessing their performance on unseen data, thereby validating the framework's overall effectiveness.

2. Materials and Methods

2.1. Overview of Methodology

The conventional paradigm for predictive machine learning, particularly in industrial diagnostics, is fundamentally reactive. This typical process relies on identifying and predicting failures from labeled datasets, a procedure that is inherently tedious and time intensive. Specifically, it necessitates the end user to await a physical failure event, label the associated data retrospectively, and then use this data to train the predictive model for future inference. A critical drawback of this reactive approach is the significant financial cost incurred by the industrial end user, often amounting to millions, even when the model successfully predicts the failure *ex post facto* (Figure 1).

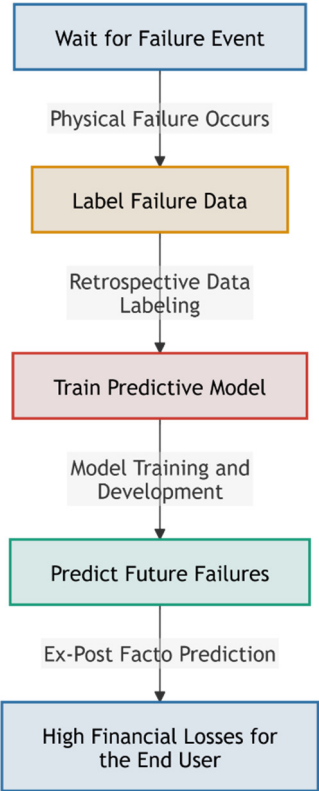


Figure 1. Reactive predictive-maintenance workflow requiring failure occurrence, retrospective labeling, and costly ex-post model training.

To mitigate this deficiency, our proposed methodology introduces a proactive mechanism. Initially, the existing operational data is subjected to clustering analysis to delineate distinct operational phases characteristic of an industrial machine. Subsequently, these clusters are semantically labeled using either domain expertise or Physics-Informed Labeling (PIL). This process establishes a dynamic human-in-the-loop framework that facilitates the provision of necessary training data at a substantially earlier stage of the machine's operation, well before the point of imminent failure. Effectively, this methodology instantiates a reinforcement learning loop that ensures the robust and continuous enrichment of the model's training dataset (Figure 2).

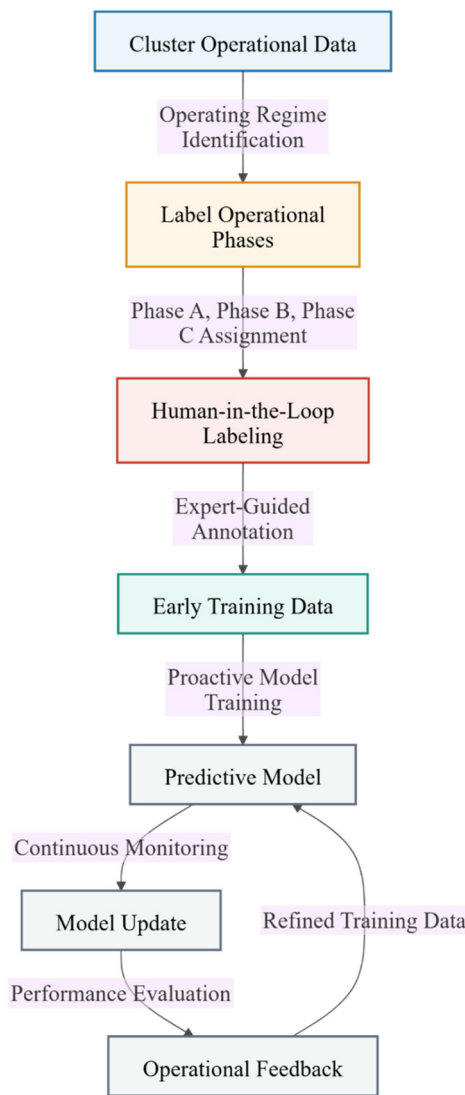


Figure 2. Proactive methodology using clustering and human/physics-informed labeling to generate early training data and enable continuous model refinement.

2.2. Dataset Characteristics and Acquisition

The empirical foundation of this research is a multivariate dataset sourced from the Kaggle repository, comprising 20,000 discrete observations derived from five distinct industrial pumps [20]. This dataset was originally curated for the analysis of mechanical health and the prediction of system failures through a range of operational parameters, including temperature, vibration intensity, pressure, flow rate, and rotational speed (RPM). Furthermore, the data incorporates accumulated operational hours and a binary maintenance flag, indicating whether intervention was historically required.

While the dataset provides a robust framework for analysis, certain metadata are absent; specifically, the precise pump configurations, sensor specifications, and measurement units were not disclosed by the original source. An exploratory data and statistical analysis conducted during the preliminary phase of this study suggests that the dataset is synthetically generated, as the maintenance flag exhibits a symmetric 50:50 distribution between normal operation and requires maintenance which is impossible in a real-life dataset where the ratio tends more towards the former. However, this is a common occurrence in predictive maintenance repositories where proprietary industrial data is restricted, and the models need equally distributed target labels for better results.

Furthermore, it is important to emphasize that the primary objective of this study is the proposition and exposition of a novel methodology for industrial pump predictive maintenance

rather than the empirical validation of a specific mechanical system. Consequently, the synthetic nature of the data does not detract from the study's utility as a proof-of-concept for the proposed framework.

Therefore, to align with the proactive approach of our methodology, the original maintenance flag was removed from the dataset. In its place, new target labels were generated through PIL following an initial clustering phase, ensuring that the model's training is driven by physical principles rather than potentially latent reactive labels.

2.3. Computational Environment and Software Specifications

To ensure experimental reproducibility and facilitate an accessible framework for replication, all simulations were executed within the standard Google Colab cloud computing environment. This Linux-based infrastructure serves as a standardized alternative to high-performance computing architecture. The framework was implemented entirely in Python within this environment, utilizing a specialized suite of open-source libraries for data manipulation, machine learning, and model explainability. Exact package versions are documented in the accompanying GitHub repository.

2.4. Methodological Framework

The proposed analytical framework for prognosticating the operational health of industrial pumping systems is structured as a systematic, multi-stage computational pipeline. This methodology synthesizes unsupervised learning for latent state discovery, physics-informed semantic annotation, and robust supervised learning for predictive diagnostics.

2.4.1. Exploratory Data Analysis (EDA) and Statistical Characterization

Following initial data integrity verifications, a rigorous statistical characterization was performed. Descriptive statistics, specifically skewness and kurtosis, were computed to evaluate the distribution and tail-behavior of numerical features. To determine the suitability of the data for unsupervised learning, the Hopkins statistic was calculated at both global and unit-specific levels to quantify the spatial randomness and inherent clusterability of the dataset. The Hopkins statistic evaluates the spatial randomness of a dataset by comparing the distances between randomly sampled points and their nearest neighbors in the real data (w_j) against the distances between generated uniform synthetic points and their nearest real neighbors (u_j):

$$H = \frac{\sum_{j=1}^m u_j}{\sum_{j=1}^m u_j + \sum_{j=1}^m w_j} \quad (1)$$

where m represents a randomly selected subset of observations ($m = 0.5 \times n$). To ensure mathematical integrity during the nearest-neighbor search, w_j was calculated using $k = 2$ neighbors to measure the distance from each selected point x_j to its nearest distinct sample point, explicitly excluding self-distance ($k = 1$, distance = 0). Under standard statistical conventions, a Hopkins Statistic near $H \sim 0.50$ indicates a spatially random or uniform distribution with no natural clustering tendency. Conversely, values approaching $H \rightarrow 1.0$ indicate highly clustered data, while $H \rightarrow 0.0$ signifies a regularly spaced grid. It is worth to mention here that, establishing baseline clusterability is essential prior to applying unsupervised state discovery algorithms, as clustering algorithms will partition uniform feature spaces regardless of whether natural boundaries exist.

To investigate the statistical homogeneity of the observations, an Analysis of Variance (ANOVA) was executed to assess if significant differences in feature means existed across distinct Pump IDs. The resulting lack of statistically significant variance is consistent with synthetic data generation, where parameters are typically sampled from uniform underlying distribution. While this homogeneity may reduce the stochastic noise characteristics of physical systems, it facilitates a controlled environment to validate the logical architecture of the proposed framework. However, it is worth mentioning here that post PIL we were able to observe meaningful variances between the identified clusters which were previously absent at the pump level aggregation.

2.4.2. Feature Engineering and Pre-Processing

To ensure numerical stability and expedite model convergence, the dataset was subjected to Z-score normalization. This process scaled all features to a distribution with a mean of 0 and a standard deviation of 1, thereby preserving the relative spacing of the original data while maintaining the true impact of potential outliers. Furthermore, an interaction term, temperature $\times$ vibration, was engineered to capture the synergistic effects of thermal and mechanical stressors on the degradation of pump components.

2.4.3. Unsupervised State Discovery and Clustering Strategy

The delineation of latent operational phases within the high-dimensional feature space was achieved through the implementation of a K-means++ clustering algorithm. To ensure the selection of a mathematically optimal cluster density (k), the elbow method was utilized, facilitated by the KElbowVisualizer utility from the YellowBrickCluster package to identify the point of diminishing returns within-cluster sum of squares. Utilizing this automated utility mitigates the subjectivity and potential inaccuracies inherent in manual heuristic identification of the inflection point.

A critical methodological distinction was drawn during this stage regarding the data aggregation strategy, specifically comparing global clustering against localized, pump-specific clustering. While a global approach, clustering the aggregate dataset, may yield higher initial performance metrics due to increased sample density, it relies on the potentially flawed assumption of homogeneous degradation trajectories across all units. Conversely, this study prioritizes individualized clustering as a more robust and ecologically valid strategy for predictive maintenance. By treating each pump as an independent entity, the framework accounts for idiosyncratic wear patterns and unit-specific operational deviations. This localized paradigm ensures higher fidelity in real-world diagnostic applications, acknowledging that even identical mechanical models manifest unique failure characteristics over time.

2.4.4. Statistical Validation and Physics-Informed Labeling (PIL)

To establish the mathematical distinctness of the identified clusters, One-Way ANOVA tests were performed across all numerical features. For features exhibiting significant variance, Tukey's Honestly Significant Difference (HSD) *post-hoc* analyses ($\alpha = 0.05$) were utilized to identify specific pairwise differences between cluster centroids.

The pivotal stage of this framework involves the transition from unsupervised clusters to semantically meaningful target variables. Utilizing Physics-Informed Labeling (PIL) and domain expertise, clusters were mapped to specific operational states: 'Healthy', 'Normal', 'Impending Failure', and 'End of Life'. These classifications were cross-referenced with the theoretical bathtub failure rate curve and Barringer-Nelson performance curves to align the data-driven clusters with established reliability engineering principles [21].

2.4.5. Predictive Modeling and Performance Evaluation

The supervised learning phase utilized an ensemble of diverse classifiers, with a particular emphasis on imbalanced learning heuristics to address the class distribution challenges inherent in the synthetic dataset. The model suite incorporated a spectrum of algorithmic approaches, ranging from stochastic and linear classifiers, such as Logistic Regression, Decision Trees, and Random Forests, to advanced gradient boosting frameworks including eXtreme Gradient Boosting (XGBoost) and Light Gradient Boosting Machine (LightGBM). To further enhance predictive stability, specialized resampling ensembles, specifically the Balanced Random Forest and Easy Ensemble, were integrated into the pipeline.

To ensure a robust and comprehensive evaluation, model performance was primarily assessed through out-of-fold predictions from Stratified 10-fold cross-validation (CV), complemented by external validation on an independent test set. We employed a multifaceted suite of evaluation

metrics derived from the multi-class confusion matrix components: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). The primary metrics utilized in this study are namely, Accuracy (ACC), Macro-averaged Recall (REC_{macro}), Macro-averaged Precision (PRE_{macro}), Macro-averaged F1-score (F1), and the Matthews Correlation Coefficient (MCC). These metrics are calculated as follows:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$PRE_{macro} = \frac{1}{N} \sum_{i=1}^N \frac{TP_i}{TP_i + FP_i} \quad (3)$$

$$REC_{macro} = \frac{1}{N} \sum_{i=1}^N \frac{TP_i}{TP_i + FN_i} \quad (4)$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (5)$$

To account for the imbalanced nature typically encountered in industrial diagnostic datasets, this study prioritized the MCC over conventional accuracy. By considering all four quadrants of the confusion matrix, the MCC ensures a balanced evaluation of model quality across all health states, regardless of their frequency in the training corpus.

2.4.6. Hyperparameter Optimization and Interpretability

To maximize the predictive efficacy of the framework, Bayesian Optimization was executed via the Hyperopt library to meticulously fine-tune the hyperparameter space of the Balanced Random Forest Classifier. This approach allowed for an efficient search of the high-dimensional parameter landscape, targeting the minimization of the negative MCC. Recognizing that high-performance models often lack transparency, the black-box nature of the final architecture was addressed through the integration of the LIME (Local Interpretable Model-agnostic Explanations) interpretability framework. LIME was utilized to generate granular, instance-level transparency by approximating the underlying model locally with an interpretable linear surrogate. By providing these localized explanations, the framework effectively translates complex algorithmic outputs into actionable maintenance insights, bridging the gap between automated diagnostics and human-centered decision-making.

3. Results

3.1. Statistical Assessment of Data Authenticity and Homogeneity

ANOVA is a statistical test used to compare the means of three or more groups. It assesses whether the statistical means of two or more groups are significantly different from each other. In the present scenario, ANOVA was conducted across the primary operational features, namely, temperature, vibration, pressure, flow rate, RPM, and operational hours, to evaluate the statistical variance between individual assets identified by Pump ID. For each parameter tested, the resulting p-values consistently exceeded the 0.05 significance threshold. These results demonstrate an absence of statistically significant differences in the mean operational values across the five distinct pumps.

In empirical industrial contexts, such a high degree of statistical uniformity is atypical. Individual mechanical units, even those of identical make and model, generally exhibit distinct performance profiles due to manufacturing tolerances, installation nuances, heterogeneous operating environments, and varying rates of component degradation. The observed lack of variance suggests that the dataset does not capture the stochastic noise and idiosyncratic behaviors inherent to physical machinery. Consequently, the stark homogeneity evidenced by the ANOVA profiles strongly confirms the synthetic origin of the dataset. The data architecture appears designed to minimize unit-

to-unit variation, likely to provide a balanced or idealized framework for algorithmic development rather than to represent the complex operational diversity found in real-world industrial systems.

3.1.1. Normality Assessment and Distributional Analysis

To further evaluate the underlying nature of the dataset, a combination of visual inspection via Quantile-Quantile (Q-Q) plots (Figure 3) and formal statistical testing using the Shapiro-Wilk test was employed.

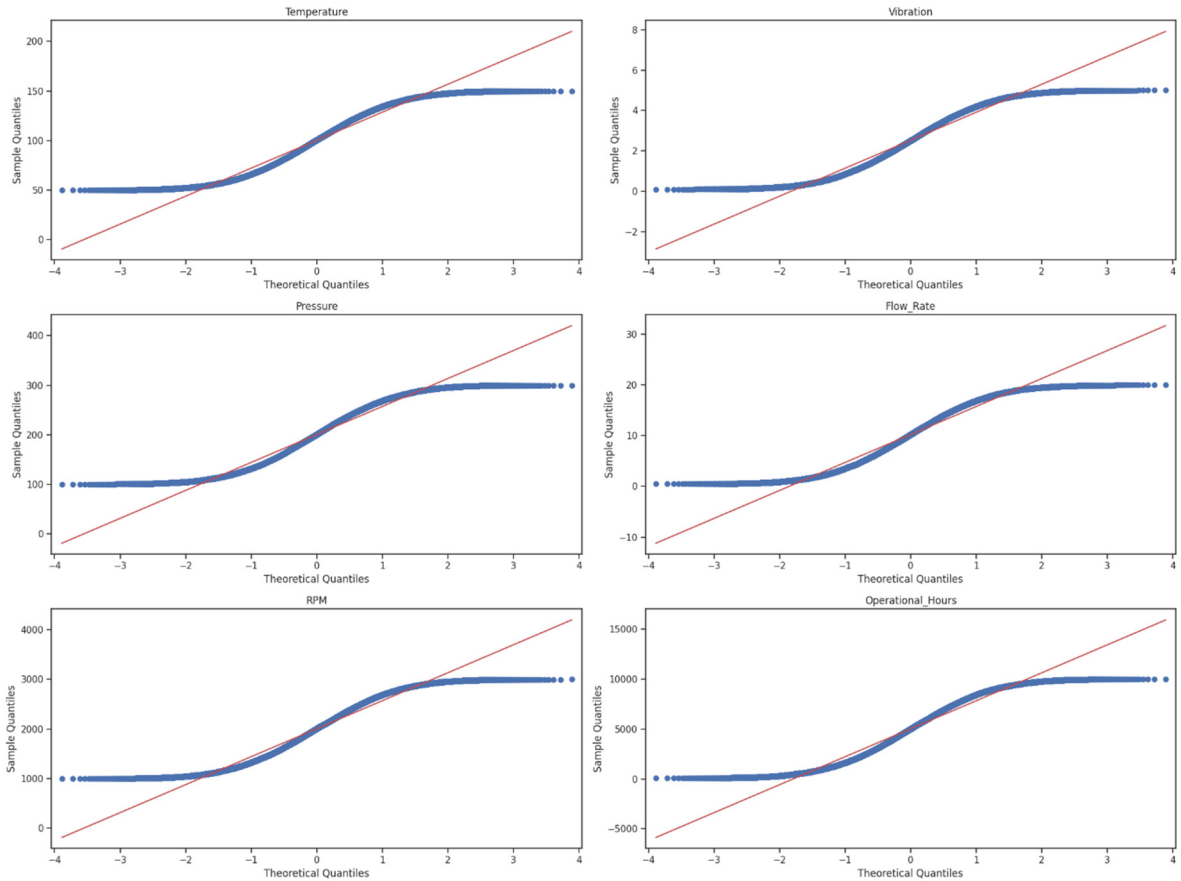


Figure 3. Q-Q plots of operational variables demonstrating a systematic S-curve deviation from normality and characteristic tail truncation.

The visual evidence provided by the Q-Q plots corroborate the statistical findings while adding significant nuance to the characterization of the dataset's non-normality. Each plot reveals a distinct S-shaped deviation from the theoretical normal line, indicating thin tails and truncation where data points flatten at both the lower and upper extremities. This suggests that the dataset lacks the long tails typically associated with stochastic noise in real-world sensors, featuring instead strictly bounded ranges. Furthermore, the striking uniformity in these deviations across all six operational variables, ranging from thermal to mechanical metrics, is highly irregular for empirical data. In physical systems, disparate phenomena such as fluid dynamics and mechanical vibration typically manifest unique distributional signatures; however, the identical curvature observed here points toward a systematic, synthetic generation process rather than independent physical origins. Furthermore, the Shapiro-Wilk test results, for all the tested variables, yielded a p-value of 0. Since these values are significantly lower than the standard alpha level ($\alpha = 0.05$), the null hypothesis is rejected for every parameter. This statistically confirms that none of the operational features follow a Gaussian distribution.

The consistent Shapiro-Wilk statistics ($W \sim 0.95$) and the uniform S-curves across disparate physical metrics further reinforce the conclusion that the dataset is synthetically generated. The

patterns further validate that the data was likely produced using a consistent mathematical transformation or clipped uniform distribution rather than being derived from independent physical processes. This systematic non-normality necessitated the implementation of robust, non-parametric machine learning algorithms, such as LightGBM and Balanced Random Forests, which do not rely on Gaussian assumptions to achieve optimal predictive performance.

3.2. Univariate Analysis and Feature Characterization

The univariate analysis provides a comprehensive statistical profile of the operational parameters, revealing a high degree of structural uniformity across the dataset. Descriptive statistics, summarized in Table 1, demonstrate that all features exhibit near-zero skewness and significantly negative kurtosis.

Table 1. Descriptive statistical summary of operational features.

Parameter	Count	Mean	Standard Deviation	Minimum	25%	50%	75%	Maximum	Skewness	Kurtosis
Temperature	20000	100.37	28.85	50.00	75.40	100.75	125.42	150.00	-0.02	-1.20
Vibration	20000	2.53	1.42	0.10	1.30	2.52	3.76	5.00	0.01	-1.20
Pressure	20000	200.38	57.82	100.01	149.86	200.51	250.27	299.98	-0.01	-1.21
Flow Rate	20000	10.25	5.65	0.50	5.35	10.28	15.15	20.00	-0.01	-1.21
RPM	20000	2005.62	576.51	1000.04	1506.07	2009.73	2503.34	2999.92	0.00	-1.19
Operational Hours	20000	5030.48	2867.80	100.21	2531.55	5032.76	7504.46	9998.77	0.01	-1.21

The visual evidence from distribution and box plots (Figure 4) corroborates these metrics, identifying two primary characteristics of data, namely absence of outliers and uniformity and platykurtic distributions.

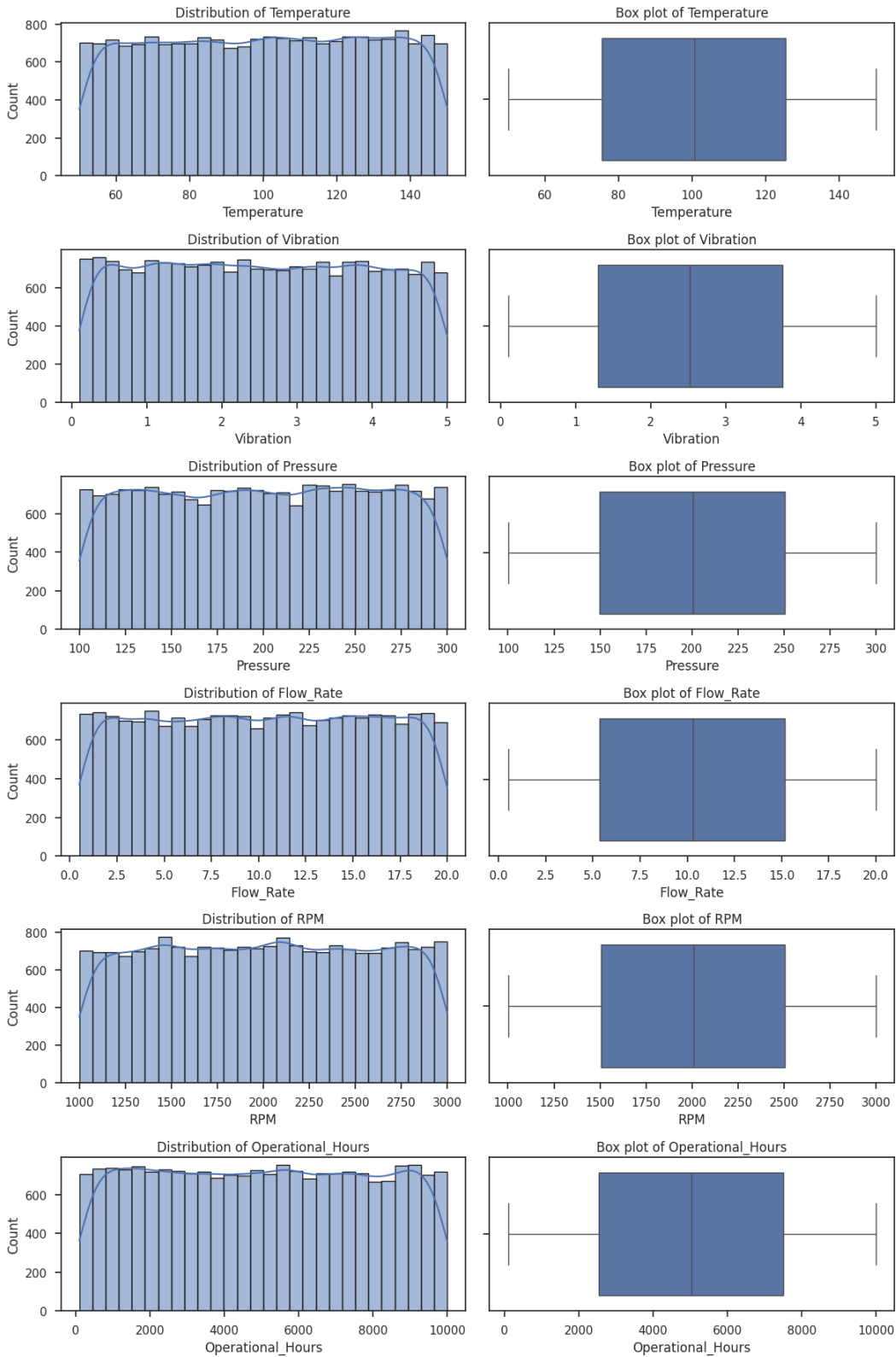


Figure 4. Distribution and box plots highlighting platykurtic profiles and the absence of statistical outliers across all sensor metrics.

All operational parameters, manifest nearly flat, rectangular probability density functions (PDFs). This is reflected in kurtosis values consistently near -1.20, signifying a platykurtic distribution that lacks the central peak and heavy tails characteristic of Gaussian processes. On the other hand, box plot analysis confirms that data points are strictly contained within established minimum and

maximum boundaries. No extreme values or statistical outliers were detected beyond the whiskers, indicating a clean, pre-processed, or synthetically bounded data range.

3.3. Bivariate Analysis and Feature Interdependency

To investigate the relationships between operational variables and identify potential multicollinearity, a comprehensive bivariate analysis was conducted using a combination of scatter plot matrices and correlation coefficients. The results, visualized in the correlation heatmaps (Figure 5) and pair plots (Figure 6), provide further insight into the structural characteristics of the dataset.

3.3.1. Correlation Matrix and Heatmap Analysis

The interdependency between operational variables was evaluated through both Pearson (linear) and Spearman (rank-based) correlation analyses to detect potential multicollinearity or monotonic trends. The resulting heatmaps (Figure 5) reveal an almost total absence of statistical relationships between the monitored parameters, with correlation coefficients for all feature pairs, such as temperature vs. vibration or pressure vs. flow rate, consistently hovering near 0.00 and failing to exceed a negligible absolute value of 0.01. This lack of correlation indicates that each feature functions as an independent dimension within the high-dimensional feature space.

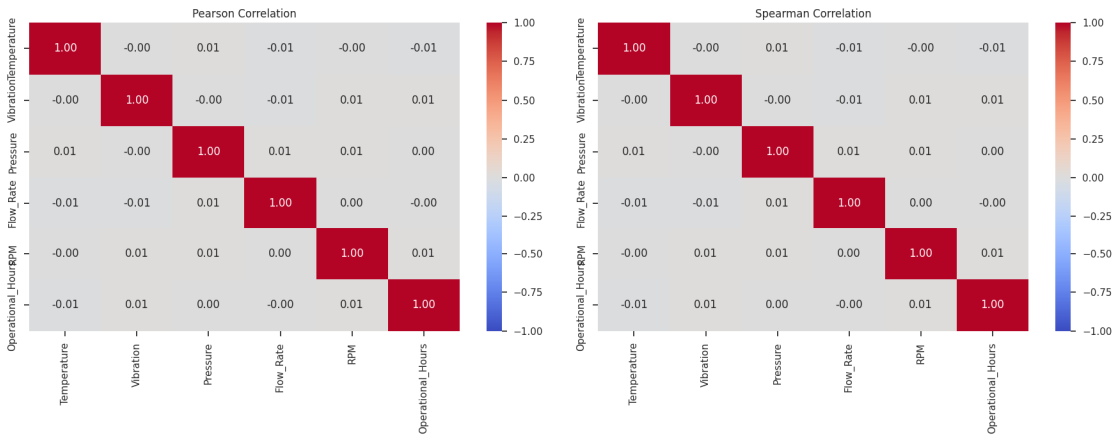


Figure 5. Pearson and Spearman correlation heatmaps displaying near-zero coefficients, confirming the lack of linear and monotonic dependencies across all features.

3.3.2. Pairwise Distribution Plots

The interdependency of the operational features was further explored through a matrix of Kernel Density Estimate (KDE) pair plots (Figure 6), which provide a sophisticated view of the dataset's high-dimensional structure. Unlike standard scatter plots, the KDE contours reveal a nuanced internal architecture characterized by localized density pockets, specific regions where data points cluster with higher frequency. While the broad, rectangular boundaries of these plots confirm the statistical independence and strict capping of the variables, these internal islands of concentration indicate that the distribution is not perfectly uniform.

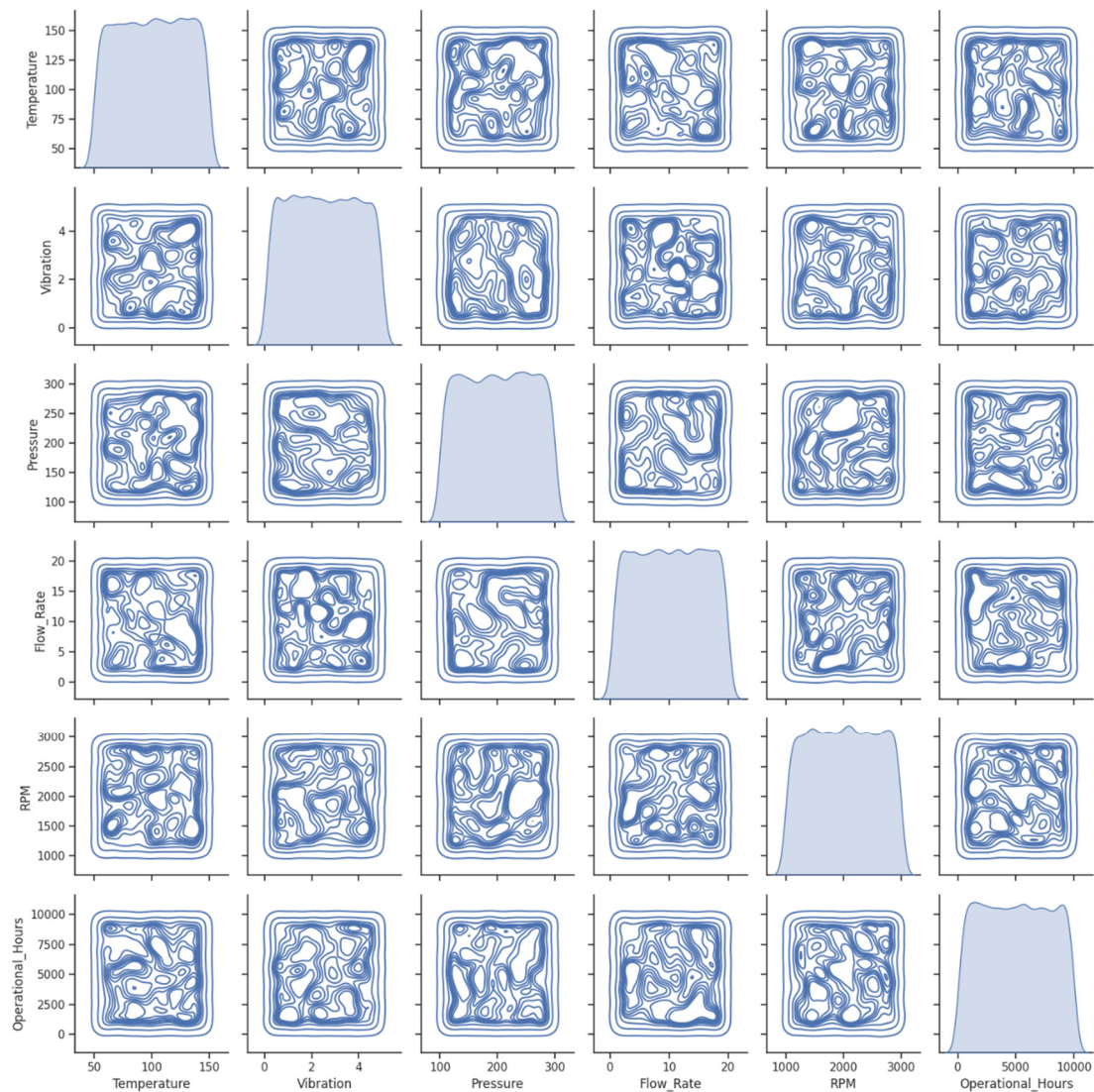


Figure 6. Multivariate pair plot illustrating the independent *cloud-like* scatter distributions and uniform marginal histograms of the operational variables.

The combination of independent features and non-Gaussian distributions identified in the univariate and bivariate stages confirms that linear diagnostic models may struggle to capture the underlying health states of the pumps. While this orthogonality mitigates the risks associated with multicollinearity in the subsequent modeling phase, the absence of inherent physical correlations, such as the characteristic positive relationship between RPM and flow rate governed by affinity laws in centrifugal systems, further corroborates the hypothesis that the dataset is synthetically generated. Specifically, the near-zero Spearman coefficients suggest that the variables were produced through independent random sampling or uncoupled mathematical transformations rather than being derived from a physically integrated process. This structural independence confirms the necessity of using non-linear, ensemble-based learners capable of partitioning the feature space along these independent axes without relying on global feature coupling.

3.3.3. Spatial Randomness and the Hopkins Statistic

Following global Z-score normalization across all six continuous operational parameters, feature distributions were evaluated individually per pump asset ($m = 2,000$ per pump, $N = 20,000$ total). Both the localized Hopkins statistics across all five pump units and the global metric consistently yielded $H \sim 0.50$. These empirical values closely track the theoretical baseline of 0.50 for uniform spatial

randomness and corroborate with the visual observations in figures 1 and 2. This strong quantitative evidence confirms that the operational dataset exhibits no naturally emergent clustering structure. In an empirical industrial setting, operational data typically aggregates into distinct high-density clusters representing steady-state operating points or specific degradation regimes. In this synthetic context, the lack of inherent spatial clustering implies that any operational states identified during subsequent partitioning will be imposed by the clustering algorithm rather than naturally emergent. This finding underscores the critical necessity of post-hoc statistical validation and PIL to ensure that discovered segments reflect physically meaningful operational regimes.

3.4. Physics Informed Labelling Through Clustering

Building upon the preceding distributional analysis (Figure 6), it is evident that despite the synthetic nature of the dataset, meaningful operational states can be derived through the integration of domain knowledge. To isolate these states, the dataset was partitioned by Pump ID, and a localized clustering approach was applied to each asset’s unique operational profile.

Using the K-means++ algorithm as the underlying clustering mechanism, the optimal number of clusters (k) for each pump was determined using the `KElbowVisualizer`, which evaluates the relationship between the number of clusters and the resulting distortion score. The optimal k is identified at the *elbow* point, the juncture where the rate of decrease in distortion significantly decelerates and the curve begins to plateau. As illustrated in Figure 7, the elbow curve for Pump 1 demonstrates a sharp inflection at $k = 4$, with a corresponding distortion score of $\sim 17,277$. Furthermore, the scatter plot in Figure 7 illustrates the identified clusters for the pressure vs. flow rate profile for the same pump.

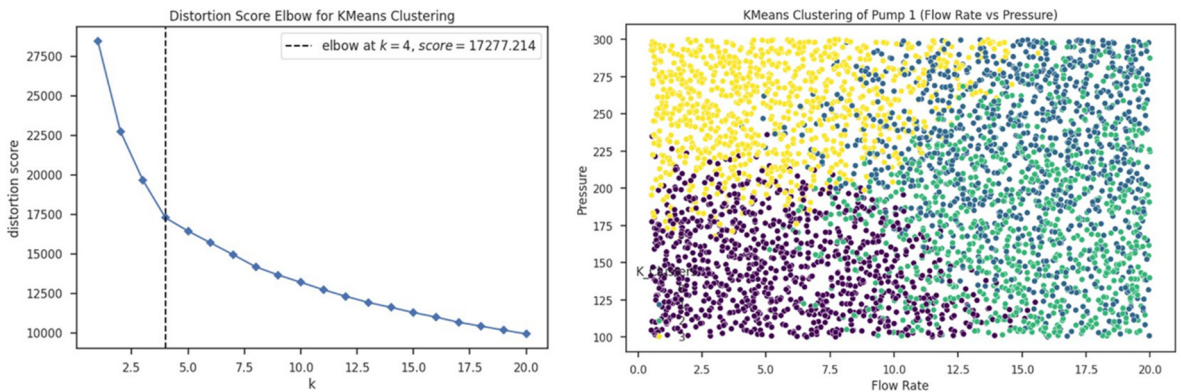


Figure 7. K-Means clustering analysis for Pump 1. (Left) Elbow method results indicating an optimal $k=4$ for feature partitioning; (Right) 2D visualization of resulting clusters mapped against pressure and flow rate.

The identification of four distinct clusters suggests that the unsupervised model successfully partitioned the feature space into four latent operational regimes. While Figure 7 specifically details the results for Pump 1, it is important to note that all other pumps in the dataset exhibited nearly identical elbow curves and distortion scores, consistently identifying $k = 4$ as the optimal cluster count with scores remaining above the 17,000 thresholds. This uniformity across the fleet reinforces the structural consistency of the dataset. Furthermore, the relatively high distortion score at the elbow point is expected; it aligns with the high spatial entropy and cloud-like KDE patterns observed previously in Figure 6, confirming that while clusters exist, they occupy a contiguous and densely packed feature space. These clusters provide the requisite structural labels for the subsequent Physics-Informed Labeling (PIL) process.

Following the identification of four optimal clusters per asset, a detailed profile analysis was conducted to map these mathematical groupings to physical operational states. By synthesizing the centroid values for each cluster with established mechanical failure curves, specifically, the Bathtub

Curve for lifecycle stages (Figure 8a) and the Barringer-Nelson reliability or the pump performance curve for health (Figure 8b), the unsupervised clusters were transformed into diagnostic labels namely, healthy, normal, impending failure and end of life as elucidated in Table 2.

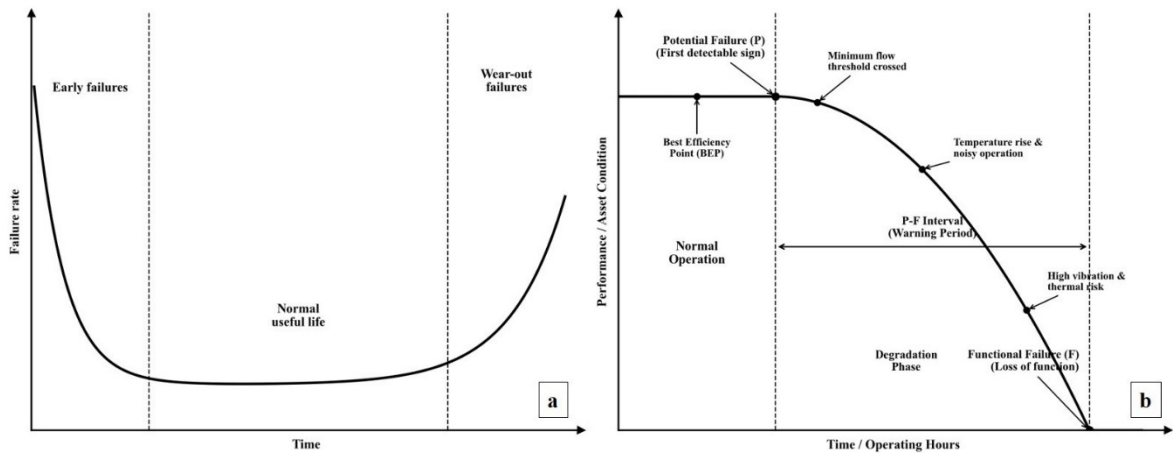


Figure 8. Conceptual framework showing the dual-model approach for diagnostic labeling, combining (a) reliability over time with (b) performance characteristics to establish ground-truth operational states.

On the other hand, Table 3 maps the identified mathematical clusters to their respective diagnostic states, *Healthy*, *Normal*, *Impending Failure*, and *End of Life*, by correlating cluster centroids with the Bathtub Failure Curve for temporal reliability and the Pump Performance Curve for health.

Table 2. Diagnostic definitions and cluster mapping for Physics-Informed Labeling (PIL). This table details the criteria for assigning operational labels based on sensor-derived metrics and lifecycle stages.

Operational Label	Physical Logic & Diagnostic Inference
Healthy	Represents the Infant stage or operation at the Best Efficiency Point (BEP). Characterized by minimal cumulative service and high stability despite high output.
Normal	Steady-state operation within design envelopes. Includes the Shut-off zone (low-flow/high-pressure) and regions where parameters remain within nominal limits.
Impending Failure	Early warning state defined by localized thermal or vibratory anomalies (> 3.0) deviating from the baseline, suggesting active mechanical degradation.
End of Life	Correlated with the wear-out phase of the Bathtub Curve. Terminal stage where cumulative fatigue increases failure probability regardless of current stability.

The results demonstrate a high degree of consistency in how the K-Means++ algorithm partitioned the feature space. A notable finding is the role of operational hours as a dominant separator for the *Healthy* and *End of Life* states, while the interaction between flow rate, pressure, and vibration differentiates *Normal* operation from *Impending Failure*.

Table 3. Physics-Informed Labels (PIL) for operational clusters identified via the K-Means++ algorithm.

Pu mp ID	Clus ter ID	Avg. Temper ature	Avg. Vibrat ion	Avg. Press ure	Av g. Flo w Rat e	Avg. RP M	Avg. Operati onal Hours	Clus ter Cou nt	Diagnostic Operational Characterist ics	Assig ned Label
1	0	98.75	2.58	147.56	5.84	2074.20	5885.28	992	Low flow, high pressure, moderate operational hours	<i>Normal</i>
	1	99.70	2.48	224.03	14.80	2042.69	7563.37	1027	High flow, high pressure, highest operational hours	<i>End of Life</i>
	2	101.54	2.50	180.13	14.74	1957.72	2362.82	950	High flow, lower pressure, lowest operational hours	<i>Healthy</i>
	3	102.65	2.61	254.21	5.61	1948.15	4047.59	942	Low flow, high pressure, moderate operational hours	<i>Normal</i>
2	0	99.14	2.54	145.91	15.09	1962.17	5289.23	975	High flow, moderate pressure, moderate operational hours	<i>Normal</i>
	1	70.83	2.57	197.46	5.91	2016.02	4718.79	1019	High flow, high pressure, highest vibration	<i>Impending Failure</i>
	2	101.58	2.45	256.73	14.76	2071.12	5004.83	1060	High flow, high pressure	<i>Healthy</i>
	3	127.15	2.58	199.21	5.37	1978.44	5148.69	1075	Low flow, high pressure, lowest vibration	<i>Normal</i>
3	0	99.79	1.20	203.10	5.55	1904.58	5032.86	975	Low-flow, high-	<i>Normal</i>

									pressure, very low vibration	
	1	103.80	2.61	256.9 0	14. 30	2005 .70	4907.33	1019	High-flow, high- pressure	Health y
	2	97.88	2.37	145.3 2	14. 99	2042 .21	4689.68	1060	High-flow, low- pressure	Norma l
	3	101.03	3.95	188.7 0	5.8 0	2065 .66	5567.57	1075	Low flow, high- pressure, highest vibration and temperature	Impen ding Failure
4	0	92.34	2.23	199.6 5	13. 89	1576 .05	7167.40	941	High flow, low-rpm, high operational_ hours	End of Life
	1	98.22	2.58	199.0 0	6.8 8	2487 .49	7065.71	1074	Low flow, high-rpm, high operational_ hours	End of Life
	2	104.31	2.53	209.5 4	14. 55	2377 .24	2786.59	978	High flow, high-rpm, lowest operational_ hours	Health y
	3	105.28	2.84	192.5 9	6.1 6	1582 .98	3145.28	1043	Low flow, low-rpm, high vibration/te mp for low operational_ hours	Impen ding Failure
5	0	102.91	2.31	220.7 5	14. 64	1921 .20	2411.41	1039	High flow, high- pressure, low operational_ hours	Health y
	1	102.08	2.49	250.3 5	6.0 0	2140 .13	6040.65	1010	Low flow, high- pressure, high operational_ hours	End of Life
	2	99.91	2.65	145.3 4	6.0 6	1969 .28	3937.07	959	Low flow, low- pressure,	Health y

								low operational_ hours	
3	96.46	2.55	181.1	14.	1953	7688.13	985	High flow, low- pressure, highest operational_ hours	End of Life
			9	88	.29				

The 3D scatter plot in Figure 9 provides a clear spatial visualization of how the PIL framework has partitioned the data across the fleet. By plotting the cluster centroids against pressure, flow rate, and operational hours, we can observe the distinct volumetric regions occupied by each mechanical state.

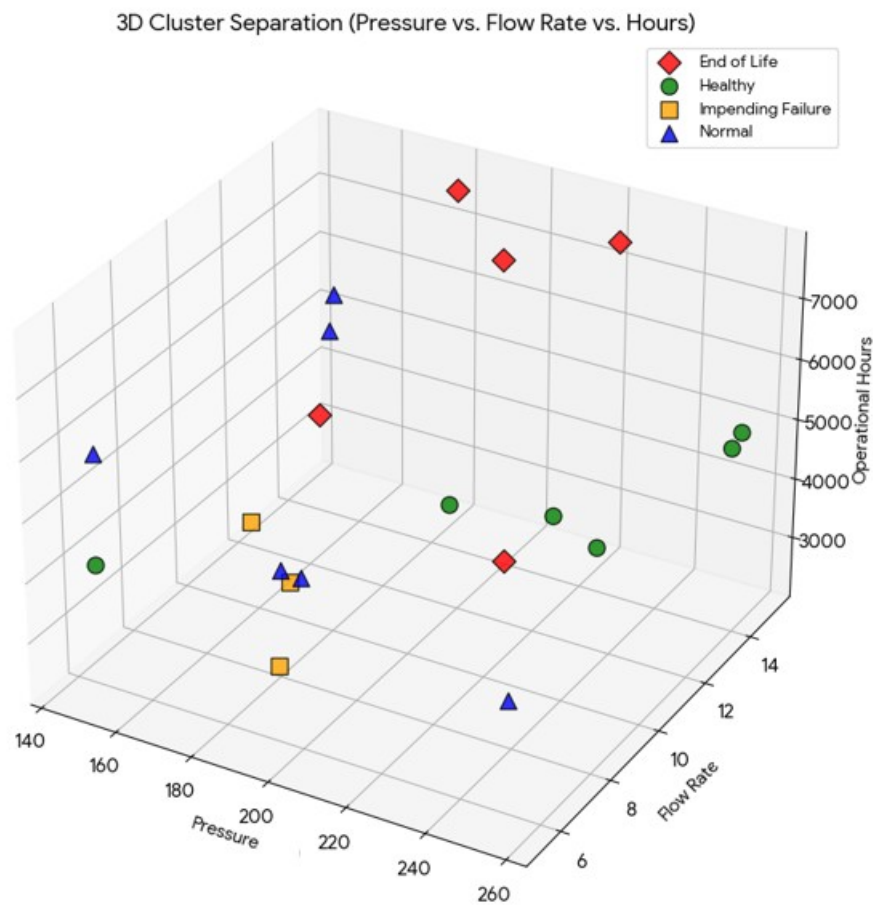


Figure 9. 3D Scatter plot showing the spatial separation of pump clusters across pressure, flow rate, and operational hours. The clear partitioning validates the use of K-Means++ centroids as the basis for diagnostic labeling.

The spatial distribution of the identified clusters validates the efficacy of the unsupervised learning framework in capturing multi-dimensional operational regimes. As illustrated in Figure 9, the feature space exhibits clear stratification along both performance and temporal dimensions. The z-axis, representing operational hours, functions as the primary differentiator for the *Healthy* and *End of Life* states. Clusters designated as *Healthy* are concentrated at the plot’s base, reflecting low cumulative service hours, whereas *End of Life* clusters occupy the upper strata, typically exceeding 6,000 hours, independent of their specific performance metrics.

Along the X (pressure) and Y (flow rate) axes, cluster distribution adheres to the inverse relationship defined by the pump performance curve. While Normal clusters are distributed across the central plane, *Impending Failure* clusters, such as those identified for Pump 2 and Pump 3, emerge as spatial outliers. These outliers are frequently situated in high-stress zones where the confluence of pressure and flow, alongside elevated vibration and temperature, deviates from the expected efficiency manifold.

Moreover, Figure 9 elucidates the underlying cause of the relatively low Hopkins statistic (~ 0.5). Although *Healthy* and *End of Life* regions remain distinct, a visible transition zone exists between *Healthy* and *Normal* states. This indicates that the degradation of asset health is a continuous progression rather than a series of discrete shifts, necessitating high-resolution boundary detection for effective predictive maintenance. However, it is pertinent to note that within a typical industrial fleet, identifying most operational states is relatively straightforward, except for *Impending Failure* signatures. Furthermore, as this study utilizes a synthetic dataset, the mean values of operational parameters and the resulting clusters exhibit high proximity; however, such demarcations are expected to be significantly more pronounced in high-fidelity, real-time industrial datasets.

3.5. Statistical Analysis of Clusters

To validate the statistical distinctiveness and diagnostic utility of the partitions generated by the K-Means++ algorithm (k=4), a nested analytical approach was employed. Beyond evaluating fleet-wide macro-trends, individual asset sub-populations (Pumps 1 through 5) were subjected to independent one-way ANOVA and pairwise Tukey’s HSD *post-hoc* testing ($\alpha = 0.05$). This unit-specific validation is essential for quantifying localized operational anomalies and testing whether macro-level boundaries hold true at the individual machine layer.

3.5.1. Statistical Analysis of Global Clusters

To confirm that the unsupervised algorithm partitioned the entire dataset into distinct operational regimes, a global one-way ANOVA was performed across the complete fleet dataset. The aggregate ANOVA results (Table 4) reveal highly significant differences ($p < 0.001$) across all six monitored engineering parameters. The exceptionally large F-statistics for vibration ($F = 13088.47$), pressure ($F = 5680.34$), and Operational Hours ($F = 5,530.26$) demonstrate that these variables possess the highest global discriminative power for defining macro-level pump health regimes.

Table 4. One-way ANOVA results for feature discriminability across operational clusters.

Operational Parameter	Degrees of Freedom	Sum of Squares	Mean Square	F-Statistic	p-value
Vibration	3	2.66×10^4	8.86×10^3	13088.47	< 0.001
Pressure	3	3.08×10^7	1.03×10^7	5680.34	
Operational Hours	3	7.46×10^{10}	2.49×10^{10}	5530.26	
Flow Rate	3	1.78×10^3	5.94×10^2	18.7	
Temperature	3	3.08×10^4	1.03×10^4	12.34	
RPM	3	1.14×10^7	3.81×10^6	11.49	

The very low p-values (< 0.001) for all features indicate that there are statistically significant differences in the means of these operational parameters across the identified clusters in the overall dataset. This supports the hypothesis that the clustering successfully segmented the data into groups with distinct operational characteristics.

To complement the global ANOVA results, a Tukey HSD *post-hoc* test was conducted. While ANOVA confirms that at least one group mean differs, the Tukey HSD test identifies exactly which pairs of clusters exhibit statistically significant differences. This granular analysis is critical for

distinguishing between closely related operational states, such as *Healthy* versus *Normal* or *Normal* versus *Impending Failure* and the results are summarized in Table 5.

Table 5. *Post-hoc* Tukey HSD results for pairwise cluster comparisons.

Operational Parameter	Cluster Comparison	Result	Null Hypothesis (H ₀)
Pressure	0 vs 1, 0 vs 2, 0 vs 3, 1 vs 2, 1 vs 3, 2 vs 3	Significant	Rejected for all pairs
Operational Hours	0 vs 1, 0 vs 2, 0 vs 3, 1 vs 2, 1 vs 3, 2 vs 3	Significant	Rejected for all pairs
Temperature	0 vs 1, 0 vs 2, 1 vs 3, 2 vs 3	Significant	Rejected
	0 vs 3, 1 vs 2	Non-Significant	Failed to Reject
Vibration	0 vs 1, 0 vs 3, 1 vs 2, 2 vs 3	Significant	Rejected
	0 vs 2, 1 vs 3	Non-Significant	Failed to Reject
Flow Rate	0 vs 2, 1 vs 2, 1 vs 3, 2 vs 3	Significant	Rejected
	0 vs 1, 0 vs 3	Non-Significant	Failed to Reject

The *post-hoc* global comparisons delineate critical structural relationships across the four-cluster topologies, demonstrating that the unsupervised clustering algorithm effectively isolates highly distinct, physics-informed operational envelopes and chronological lifecycle phases. This robust differentiation is anchored by pressure and operational hours, both of which exhibit absolute metric boundaries with an overwhelming rejection of the null hypothesis across every pairwise permutation. Specifically, flawless physical step-changes establish total pressure zone discriminability across all configurations, reaching a maximum deviation of $\Delta\mu = 109.20$ between Clusters 0 and 2, while complete statistical isolation across operational hours documents clear macro-level engine wear progression and age tracking over extended operating periods.

Conversely, the multi-tiered behaviors observed in temperature, vibration, and flow rate highlight critical machine states characterized by localized physical decoupling where specific parameters stabilize or blend. For instance, while temperature displays robust thermal boundary separation during major thermodynamic shifts (such as Clusters 2 vs 3, $p = 0$), distinct thermal stabilization plateaus occur exclusively during the transitions between Clusters 0 vs 3 ($p = 0.564$) and Clusters 1 vs 2 ($p = 0.385$). Similarly, strong mechanical step-changes identify distinct structural boundaries for vibration, yet alternative baseline profiles demonstrate clear harmonic parity between Clusters 0 vs 2 ($p = 0.707$) and Clusters 1 vs 3 ($p = 0.683$), proving that structural overlaps persist even under significantly varying thresholds. Finally, fluidic shifting patterns for flow rate remain highly distinct across core operating domains, blending only during specialized intermediate boundaries such as Clusters 0 vs 1 ($p = 0.275$) and Clusters 0 vs 3 ($p = 0.063$). Together, these narrow, overlapping physical regimes underscore the necessity of synthesizing secondary thermal and structural indicators alongside primary metrics to achieve complete, unambiguous diagnostic separation.

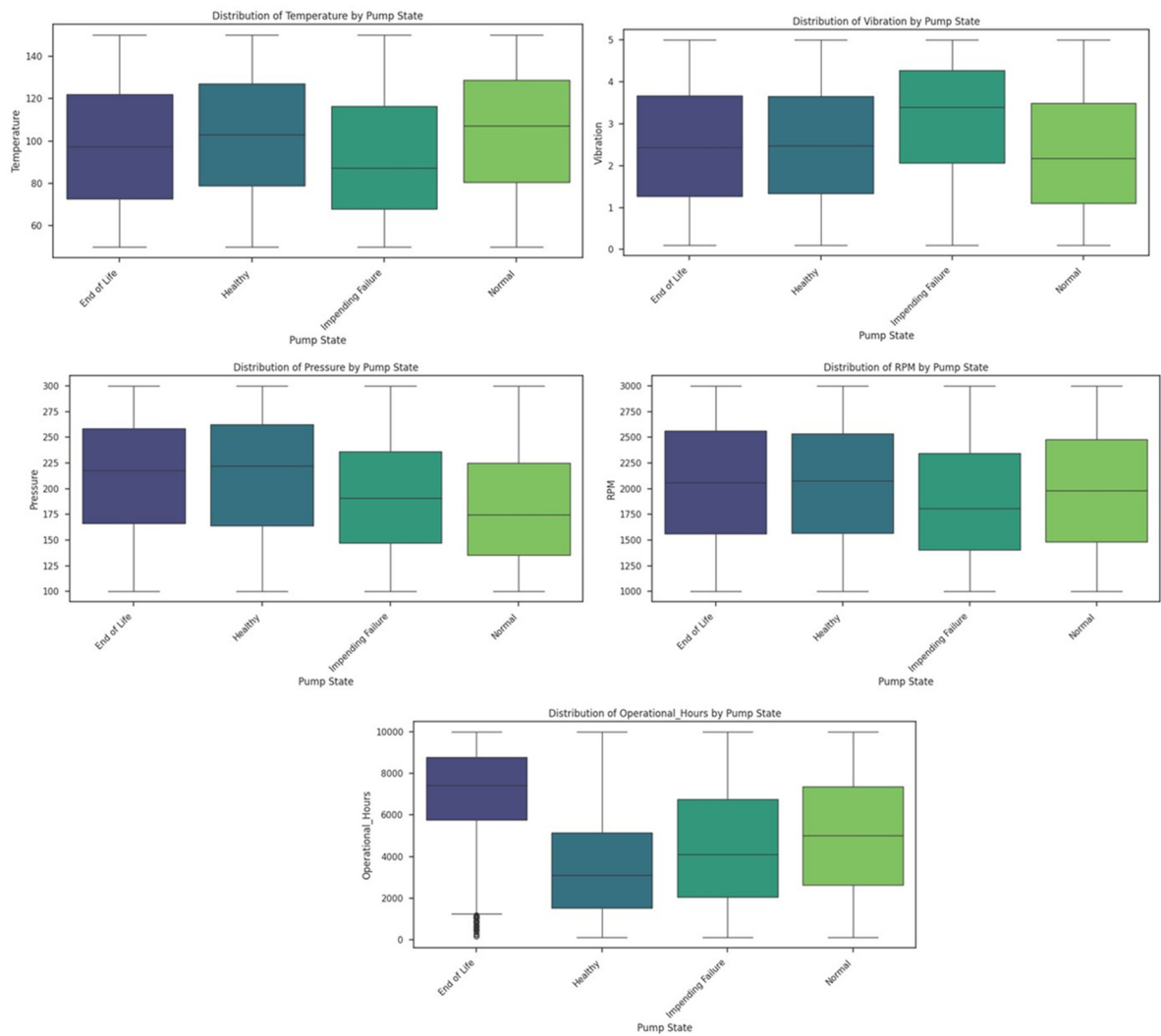


Figure 10. Global distributions of monitored parameters across the four operational pump states.

The empirical distributions visualized in the box plots in Figure 10 confirm the physiological validity of the four clustered pump states, matching the global ANOVA and *post-hoc* statistical partitions. The clear segregation in operational hours provides a strong baseline validation for chronological lifecycle tracking. Specifically, the *End of Life* state isolates an advanced runtime demographic, exhibiting a highly elevated median centered near 7,500 hours alongside a compressed interquartile range (IQR) and distinct downward outliers, which marks the cumulative degradation threshold of the fleet. In contrast, the *Healthy* state captures the early asset lifecycle, characterized by a lower median runtime below 4,000 hours. The *Normal* and *Impending Failure* regimes display broad, overlapping operational age bands, indicating that transition into a fault state is driven by acute physical stress rather than sheer historical runtime alone.

This stress-driven degradation is explicitly illuminated by vibration and temperature distributions. The *Impending Failure* regime displays a prominent upward shift in mechanical vibration, yielding the highest median score (~3.4) and a compressed upper quartile, signaling structural anomalies, severe unbalance, or localized friction. Crucially, this mechanical spike coincides with a notable drop in median temperature and lower operating pressures. This indicates a physical decoupling where the asset exhibits severe mechanical distress under lower demand profiles.

Conversely, the *Healthy* and *Normal* states operate under standard mechanical baselines but show elevated thermal profiles, capturing high-load, high-efficiency operating envelopes. The *End of Life* state reflects a combined wear signature, where moderate-to-high vibration levels persist despite

reduced performance, validating the multi-tiered physical stabilization plateaus and decoupling phenomena identified in the *post-hoc* pairwise analysis.

3.5.2. Statistical Analysis of Pump Specific Clusters

To supplement the aggregate global results, individual asset-level diagnostics were conducted across the five discrete pumps within the fleet. While the aggregate model establishes macro-level patterns, individual asset verification is required to capture local anomalies, variations in wear trajectories, and distinct environmental dependencies.

When deconstructed at the individual asset level, the fleet exhibits highly localized operational and degradation signatures that supplement macro-level global trends. Table 6 summarizes the primary variance features, key ANOVA metrics, Tukey's HSD cluster shifts, and corresponding physical interpretations for each asset.

Table 6. Asset-Level Cluster Signatures and Statistical Diagnostics across the Fleet.

Asset	Primary Variance Feature	Key ANOVA Metric	Dominant Cluster Shift (Tukey HSD $\Delta\mu$)	Primary Physical Interpretation
Pump 1	Operating Hours	$p = 0.1256$ (Vibration)	Cluster 1 (+1,678 hrs), Cluster 2 (-3,522 hrs)	Chronological Wear & Lifecycle Partitioning
Pump 2	Temperature	$F = 1213.76, p < 0.001$	Cluster 1 (-28.3°C), Cluster 3 (+28.0°C)	Cyclical Process & Thermal Sensitivity
Pump 3	Vibration	$p < 0.001$ (All pairs)	Cluster 3 ($\Delta\mu = +2.74$ mm/s Vibration)	Load-Dependent Mechanical Imbalance
Pump 4	Rotational Speed (RPM)	$p < 0.001$ (Omnibus)	Cluster 1 (+911 RPM), Cluster 2 (+801 RPM)	Dynamic Throttling & Fluid Dynamic Shifts
Pump 5	Operating Hours & Pressure	$p < 0.001$	Cluster 3 (+5,276 hrs, Pressure $\Delta\mu = -39.56$ bar)	Late-Stage Degradation & System Loss

The operational profile of Pump 1 is characterized by high mechanical stability and clear lifecycle partitioning. Because mechanical vibration remains statistically uniform across all assigned cluster boundaries ($p = 0.1256$) and thermal shifts are confined strictly to the operational limits between Cluster 0 and Cluster 3 ($\Delta\mu = +3.90^\circ\text{C}$, $p = 0.0168$), the underlying cluster assignments are heavily dictated by chronological runtime. This finding is supported by post-hoc Tukey's HSD evaluations, which confirm substantial shifts in operational history, isolating an advanced operating phase in Cluster 1 (+1,678 hours relative to Cluster 0) and an early historical baseline in Cluster 2 (-3,522 hours relative to Cluster 0).

In contrast to the long-term wear tracking observed in Pump 1, Pump 2 isolates immediate, cyclical process transitions. The variance in accumulated operational hours across its cluster boundaries is marginal, ranging between only 140 hours and 570 hours, indicating that the asset is instead governed by acute thermal sensitivity ($F = 1213.76$, $p < 0.001$), with all pairwise cluster combinations rejecting the null hypothesis. Within this thermodynamic regime, Cluster 1 captures a low-temperature operational state ($\Delta\mu = -28.3^\circ\text{C}$ relative to Cluster 0), while Cluster 3 segments a highly elevated thermal plateau ($\Delta\mu = +28.0^\circ\text{C}$ relative to Cluster 0).

Shifting from thermal to structural variation, Pump 3 presents a distinct statistical signature of load-dependent mechanical stress, distinguishing itself as the only asset in the fleet where mechanical vibration varies significantly ($p < 0.001$) across every pairwise cluster combination. Specifically, Cluster 3 isolates a severe operating envelope characterized by simultaneously elevated pressure, high rotational velocity (RPM), and a marked increase in mean vibration ($\Delta\mu = +2.74$ mm/s relative to Cluster 0), providing clear empirical evidence of mechanical imbalance or localized structural friction under peak loads.

Dynamic process adjustments are further pronounced in Pump 4, whose statistical profile is governed by high-velocity dynamic throttling. The omnibus ANOVA test reveals extreme adjustments in rotational speed, and Tukey's HSD identifies significant pairwise differences wherein Cluster 1 and Cluster 2 exhibit substantial velocity increases of +911 RPM and +801 RPM relative to Cluster 0, respectively. These substantial velocity adjustments alter the localized fluid dynamics of the system, driving concurrent, statistically significant downstream shifts across temperature, pressure, flow rate, and vibration.

Finally, Pump 5 exemplifies late-stage lifecycle divergence, where the clustering framework identifies extensive historical tracking variations across operational hours. Cluster 3 isolates a highly advanced runtime state (+5,276 hours compared to Cluster 0) that strongly correlates with distinct reductions in system pressure ($\Delta\mu = -39.56$ bar) and marked shifts in thermal equilibrium, thereby providing an empirical baseline for asset degradation and predictive maintenance modeling.

3.6. Machine Learning Modeling and Comparative Analysis

Following the statistical validation of the identified operational clusters, a comparative modelling phase was initiated to develop a robust supervised classification framework. The diagnostic states established through PIL and integrating the characteristic curves, served as the ground truth for training eight distinct architectures. These models, ranging from traditional tree-based classifiers to advanced ensemble methods, were tasked with predicting the four discrete operational regimes.

To ensure generalized performance across the synthetic fleet and mitigate potential overfitting, the models were evaluated using 10-fold Stratified CV alongside Out-of-Fold (OOF) performance metrics. Because the operational states are disparately distributed across the fleet, the evaluation criteria prioritized the MCC and Macro F1-score. The MCC was selected as the primary benchmarking metric due to its mathematical resilience against class prevalence asymmetry, providing an unbiased assessment of multi-class classification performance compared to raw accuracy.

Table 7. Performance comparison of machine learning architectures across CV and OOF sets.

Model	CV Accura cy	CV Macro Precisi on	CV Macr o Reca ll	CV Macr o F1	CV MC C	OOF Accura cy	OOF Macro Precisi on	OOF Macr o Reca ll	OOF Macr o F1	OOF MC C
Balanced Random Forest	0.5288	0.5231	0.5347	0.5234	0.3701	0.5288	0.5228	0.5347	0.5234	0.37
LightGBM	0.5354	0.527	0.5183	0.5213	0.3667	0.5354	0.5267	0.5183	0.5215	0.3666
Random Forest	0.5319	0.522	0.5118	0.5153	0.3607	0.5319	0.5219	0.5118	0.5155	0.3606
Logistic Regressi on	0.5291	0.5175	0.511	0.5121	0.3569	0.5291	0.5176	0.511	0.5124	0.3568
XGBoost	0.5254	0.5149	0.5079	0.5104	0.3532	0.5254	0.5147	0.5079	0.5106	0.3531
Easy Ensembl e	0.4997	0.4935	0.5105	0.4924	0.3355	0.4997	0.4931	0.5105	0.4927	0.3353
Balanced Bagging	0.4989	0.489	0.4964	0.4897	0.3255	0.4989	0.4885	0.4964	0.4897	0.3253

Decision Tree	0.4573	0.4419	0.442	0.441	0.262	0.4573	0.4416	0.442	0.441	0.262
				6	3				7	2

During baseline screening (Table 7), gradient boosting and class-balanced ensemble frameworks demonstrated high stability across cross-validation folds. LightGBM achieved the leading baseline CV Accuracy (0.5354) and Macro Precision (0.5270), demonstrating a strong capacity to capture complex feature interactions across operational boundaries. Meanwhile, BRF yielded slightly higher baseline recall (0.5347) due to its internal majority-class down-sampling mechanism.

To maximize predictive performance across the non-linear operational feature space, a Bayesian hyperparameter optimization phase was conducted using the Hyperopt framework powered by the Tree-structured Parzen Estimator (TPE) algorithm. Unlike traditional random or grid searches, Bayesian optimization sequentially constructs a probabilistic surrogate model of the objective function to efficiently navigate high-dimensional parameter spaces.

The TPE algorithm evaluated a 9-dimensional search space encompassing structural tree capacity. To maintain computational integrity and prevent data leakage during optimization, feature standardization was encapsulated within an imbalanced learn processing pipeline inside each cross-validation fold. The objective function directly minimized the negative mean 10-fold stratified CV MCC, explicitly aligning hyperparameter selection with the primary evaluation metric.

The comparative evaluation on the independent test set conclusively established LightGBM as the superior architecture (Table 8). Following Bayesian optimization, LightGBM successfully mapped the complex, non-linear thresholds separating target health states, achieving the strongest overall alignment with physical ground truths.

Table 8. Final test set performance metrics for optimized models.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	MCC
LightGBM	0.5445	0.5352	0.5229	0.5271	0.3774
Logistic Regression	0.5378	0.5227	0.5164	0.5168	0.3685
Balanced Random Forest	0.5233	0.5156	0.5256	0.5168	0.3606
Random Forest	0.5315	0.5201	0.5072	0.5112	0.3589
XGBoost	0.5255	0.5115	0.5036	0.5065	0.3519
Easy Ensemble	0.5088	0.4972	0.5090	0.4967	0.3440
Balanced Bagging	0.5045	0.4922	0.4991	0.4939	0.3304
Decision Tree	0.4553	0.4379	0.4390	0.4383	0.2593

LightGBM's leaf-wise tree growth strategy and histogram-based feature binning enable it to construct precise decision boundaries along continuous thermal and mechanical parameter gradients. The confusion matrix for the optimized LightGBM classifier (Figure 11) confirms exceptionally high classification fidelity for both the *End of Life* and *Healthy* operational regimes. This performance is strongly supported by the discriminative power of cumulative operational feature trajectories, such as total operating hours.

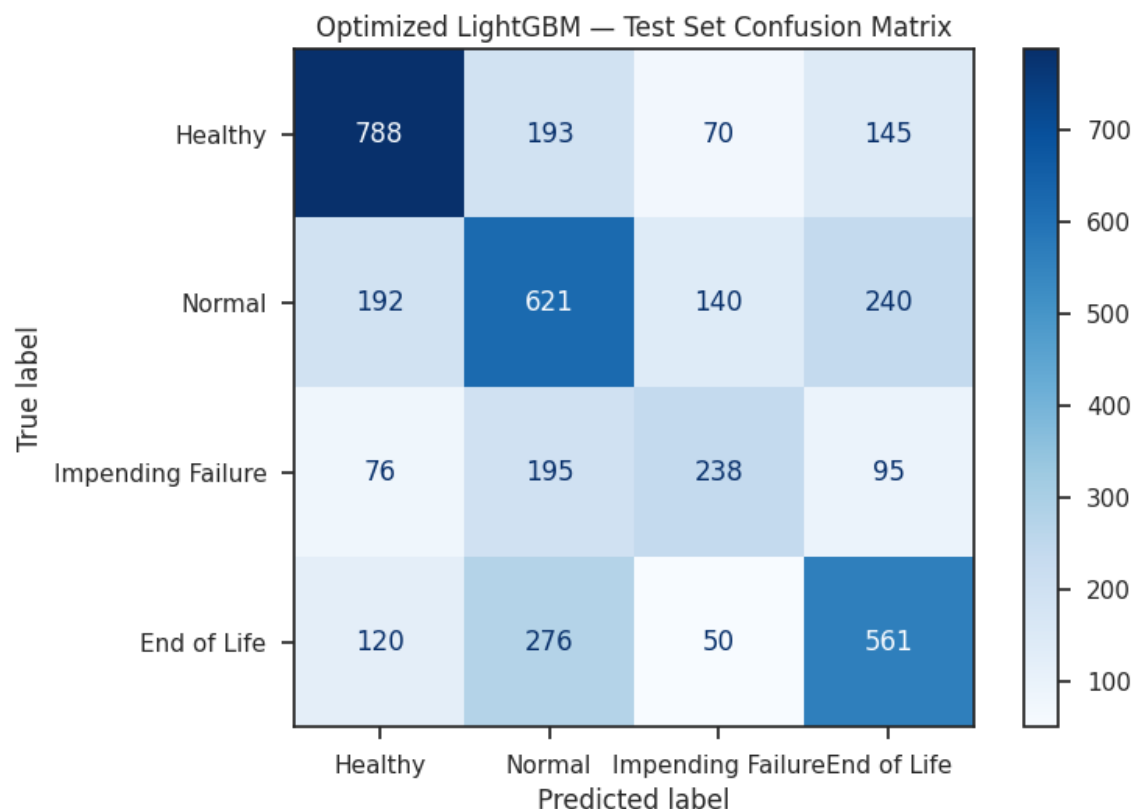


Figure 11. Confusion matrix for the optimized LightGBM classifier on the test set. Diagonal elements demonstrate robust performance across asymmetrical classes, highlighting strong discrimination for boundary states and a localized transition locus between overlapping Normal and Impending Failure regimes. .

Conversely, minor misclassifications primarily occur at the intersection of the *Normal* and *Impending Failure* boundaries. Rather than reflecting an algorithmic limitation, this structural behavior corresponds to a continuous physical transition zone: as an asset begins to undergo early mechanical degradation, its operational and thermal signatures closely mimic standard high-load steady-state profiles. By effectively modeling these subtle gradient shifts, the LightGBM framework establishes a highly reliable early-warning classification tool capable of detecting developing asset anomalies well ahead of catastrophic physical failure.

3.7. Post-Hoc Interpretability via LIME

To understand the inherently opaque *black-box* decision dynamics of the ensemble framework, Local Interpretable Model-agnostic Explanations (LIME) were deployed to provide granular *post-hoc* interpretability. By systematically perturbing individual data points in the feature space and tracking the corresponding variations in class probabilities, LIME constructs local linear surrogate models. This mathematical formulation isolates the exact influence of individual operational parameters for specific asset predictions, establishing instance-level transparency.

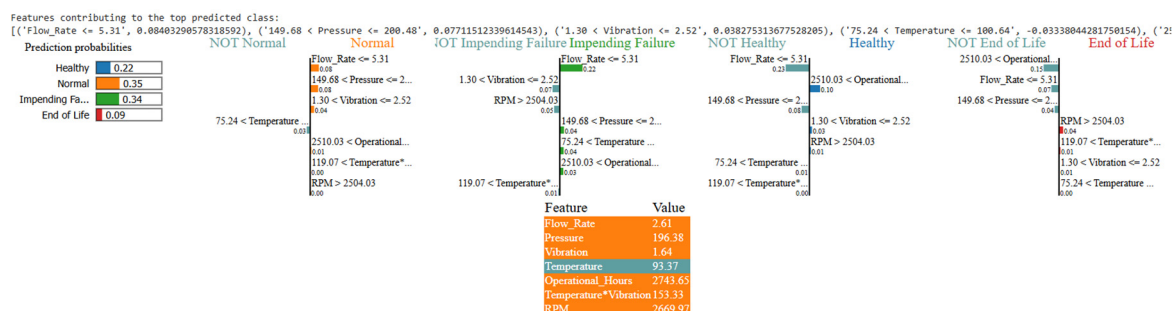


Figure 12. Screenshot of the LIME interpretation for an instance classified as *Impending Failure* by the LGBM model.

When evaluated on an asset instance classified under the *Impending Failure* regime, the local surrogate analysis identified mechanical vibration and thermal signals as the primary drivers driving the prediction (Figure 12). Crucially, the analysis revealed that even when an asset possesses moderate accumulated operational hours, an anomalous, acute spike in the vibratory signature mathematically overrides the chronological and temporal features. This parameter coupling forces the model to shift its prediction vector away from a steady-state *Normal* designation.

Conversely, for instances classified as *Healthy*, the model's decision boundary is heavily anchored by the convergence of low total operational runtimes and optimized flow rates aligned with the pump's BEP.

Integrating this interpretability layer is vital for actionable industrial deployment. It equips reliability and maintenance engineers with the diagnostic context required to look past raw categorical labels and isolate the explicit sensor triggers, such as a specific 3.95 vibration peak, that justify a physical field inspection. By bridging the gap between high-dimensional algorithmic outputs and observable thermodynamic or structural phenomena, this framework reinforces a rigorous, *human-in-the-loop* predictive maintenance paradigm.

4. Limitations of the Present Approach

While the developed framework successfully integrates physics-informed labelling with a class-balanced ensemble architecture, certain structural limitations remain inherent to its current implementation. The diagnostic framework evaluates sensor telemetry strictly as static classification snapshots, meaning that because the model lacks sequential memory, it cannot dynamically capture the continuous, time-dependent rate of change or temporal trends leading up to an operational state transition. Furthermore, the proximity of boundaries between the *Normal* and *Impending Failure* operational states introduces a localized region of high variance; the highly overlapping physical signatures in these transition zones constrain the absolute precision of the classifier, occasionally leading to false positives during high-load steady-state operations. This is compounded by deterministic output constraints, as the model provides discrete classification labels rather than continuous probabilistic confidence scores, thereby limiting its integration into risk-based decision-making matrices where plant operators must gauge underlying mathematical uncertainty for borderline instances. Finally, the baseline architecture relies on fleet-wide assumption homogeneity, assuming a uniform operational manifold across the entire pump population without explicitly accounting for minor manufacturing tolerances, unit-specific mechanical imperfections, or disparate localized installation environments that might skew an individual asset's baseline signature.

5. Directions for Future Research

To expand upon this foundation, several strategic avenues for future scholarly inquiry exist to enhance the granular accuracy, deployment capabilities, and predictive reliability of the framework. A critical progression involves transitioning from static classification snapshots to dynamic temporal architectures by utilizing recurrent networks, such as Long Short-Term Memory (LSTM) networks or Gated Recurrent Units (GRUs), which will enable the system to model stochastic state transitions over time and explicitly prognosticate Remaining Useful Life (RUL). To mitigate fleet-wide assumption homogeneity, future work will investigate localized per-pump models using an asset-specific *Digital Twin* approach, allowing the predictive pipeline to better isolate unique baseline anomalies from broader fleet shifts. Furthermore, the framework can be expanded into a probabilistic health assessment framework by employing Bayesian neural networks or Gaussian process classifiers, which will output continuous predictive distributions rather than discrete labels, providing confidence intervals to better support risk-based maintenance scheduling. Granular early-warning capabilities can also be augmented by implementing secondary unsupervised anomaly

detection layers, such as isolation forests or deep autoencoders, specifically within the established *Normal* clusters to detect subtle, sub-state deviations signifying the earliest latent stages of mechanical degradation. Finally, a vital empirical extension involves conducting a longitudinal techno-economic validation study to build formal cost-benefit models, allowing operators to explicitly measure the real-world reduction in Total Cost of Ownership (TCO), the optimization of spare-part logistics, and the financial mitigation of secondary damage costs associated with catastrophic pump failures.

Conclusion

This research successfully developed a physics-informed, class-balanced predictive framework for autonomous pump health diagnostics across an industrial fleet. By partitioning multi-sensor telemetry using a nested approach, the framework mapped complex data to physical states, by integrating the bathtub failure and pump performance curves. Statistical validation confirmed these regimes represent distinct mechanical and thermodynamic signatures. While accumulated run-time governed lifecycle baseline shifts, the localized interactions between vibration, temperature, and pressure proved vital for isolating early-stage mechanical anomalies. During supervised classification, the LightGBM framework emerged as the champion model. Optimized via Bayesian hyperparameter tuning, the gradient-boosted architecture successfully countered class prevalence asymmetry through leaf-wise tree growth and regularization, yielding high diagnostic sensitivity for the critical *Impending Failure* minority state without being masked by steady-state majority profiles. Ultimately, this integration of physics-informed labeling and Bayesian-optimized machine learning bridges the gap between raw fleet telemetry and actionable diagnostic decisions, offering a highly scalable blueprint for enhancing the operational resilience of mission-critical industrial infrastructure.

Supplementary Materials: The following supporting information can be downloaded at the website of this paper posted on Preprints.org. The complete software code, associated datasets, and further detailed information are accessible in the project repository located at: https://github.com/anatarajank/predictive_maintenance. This repository is made available under the terms of the Creative Commons Attribution 4.0 International License (CC-BY-4.0 International), which permits the unrestricted use, distribution, and adaptation of the work, provided that appropriate credit is given to the original authors.

Author Contributions: Conceptualization, A.N.; methodology, A.N.; software, A.N and A.K.A.; data curation, A.K.A.; formal analysis, A.N. and A.K.A.; validation, A.N. and A.K.A.; writing—original draft preparation, A.N.; writing—review and editing, A.N. and A.K.A.; visualization, A.N.; supervision, A.N.; project administration, A.N. All authors have read and agreed to the published version of the manuscript.

Data Availability Statement: The original data used in the present study is licensed under the MIT license and can be downloaded from here: <https://www.kaggle.com/datasets/selonamaris/large-industrial-pump-maintenance-dataset>.

Conflicts of Interest: The authors declare no conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

ACC	Accuracy
AI	Artificial Intelligence
ANOVA	Analysis of Variance
BEP	Best Efficiency Point
BRF	Balanced Random Forest
CNN	Convolutional Neural Network
CV	Cross-Validation

DL	Deep Learning
EDA	Exploratory Data Analysis
F1	F1-Score
FN	False Negatives
FP	False Positives
GRU	Gated Recurrent Unit
HSD	Honestly Significant Difference
IIoT	Industrial Internet of Things
IQR	Interquartile Range
KDE	Kernel Density Estimate
LSTM	Long Short-Term Memory
MCC	Matthews Correlation Coefficient
MCSA	Motor Current Signature Analysis
ML	Machine Learning
OOF	Out-of-Fold
PDF	Probability Density Function
PIL	Physics-Informed Labeling
PREmacro	Precision / Macro-averaged Precision
Q-Q	Quantile-Quantile
RECMacro	Recall / Macro-averaged Recall
RNN	Recurrent Neural Network
RPM	Revolutions Per Minute
RUL	Remaining Useful Life
SVM	Support Vector Machine
TCO	Total Cost of Ownership
TN	True Negatives
TP	True Positives
XAI	Explainable Artificial Intelligence

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